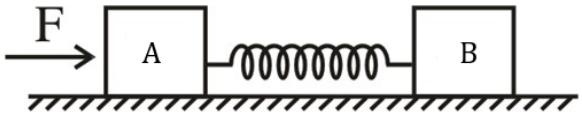
PART 1 : PHYSICS

SECTION-I (i)

1) A constant force F is pushing the block A of mass 5kg which is connected with block B of mass 10 kg by a light spring till the block B will slide. If 0.4 and 0.5 are the coefficient of friction between A and ground and B and ground respectively. (where k = stiffness of the spring)



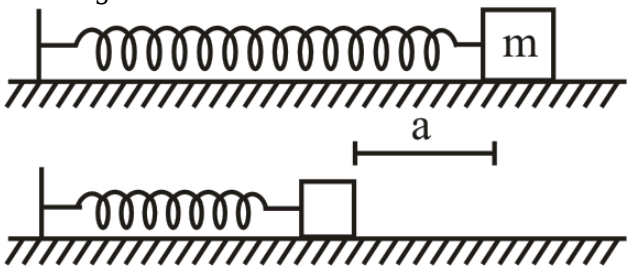
(A) The maximum compression of the spring is equal to $\frac{70}{k}$

(B) The minimum magnitude of F to move the block B is 45N

(C) The minimum magnitude of F to move the block B is 70N

(D) At maximum compression force F equals force of spring.

2) A block mass m is attached to a spring as shown in the figure. Block is displaced such that spring is compressed by distance a. Now the block is released and it comes to rest when spring comes in natural length. If coefficient of friction between block and ground is μ. Then force constant of spring is :-



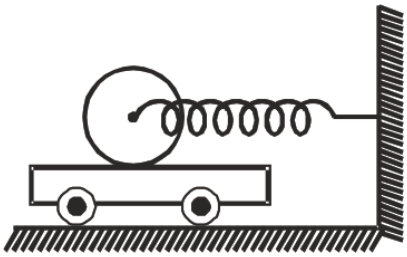
(A) $\frac{2\mu mg}{a}$

(B) $\frac{\mu mg}{a}$

(C) $\frac{\mu mg}{2a}$

(D) None

3) In the figure shown, the mass of the disc as well as that of the trolley is M. The spring is ideal and has stiffness k. The trolley can move horizontally on the floor without friction, and the disc can roll on the trolley surface without slipping. The spring is compressed and the system released so that

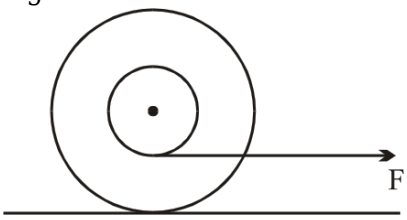
oscillations begin. The :- 

(A) Acceleration of center of disc = twice of that of trolley

(B) Acceleration of center of disc = thrice of that of trolley

(C) Acceleration of center of disc = half of that of trolley

(D) Acceleration of center of disc = that of trolley

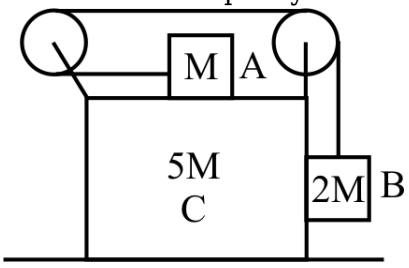
4) Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and placed over a rough horizontal surface. Thread is pulled by a force F as shown in figure, then in case of pure rolling 

(A) thread unwinds, spool rotates anticlockwise and friction act leftwards.

(B) thread winds, spool rotates clockwise and friction acts leftwards.

(C) thread winds, spool moves to the right and friction act rightwards.

(D) thread winds, spool moves to the right and friction does not come into existence.

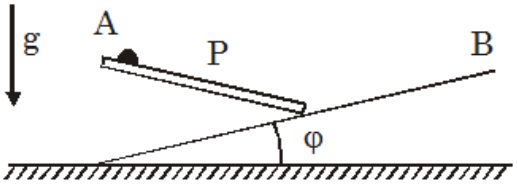
5) In the system shown the pulleys are ideal and the cords are inextensible (there is no friction anywhere) 

(A) magnitude of acceleration of block A and B will be same

(B) magnitude of acceleration of block B and wedge C will be equal

(C) magnitude of acceleration of block A will be greater then magnitude of acceleration of wedge C

(D) magnitude of acceleration of wedge C will be greater than magnitude of acceleration of block B

6) A ball is held at a fixed height from a smooth incline plane 'B' of inclination $\varphi = 60°$. Now, a smooth platform 'P' of variable length is placed below the ball and the ball is released from rest. The ball slides on the platform and touches the incline 'B' in the shortest time? Find the angle of platform 'P' with vertical (in degrees). 

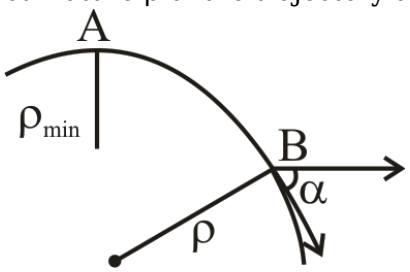
(A) 0

(B) 90

(C) 30

(D) 45

7) If ρ_{min} is radius of curvature of trajectory of a projectile at highest point A, then the radius of curvature ρ of the trajectory at B where tangent on it forms α angle with horizontal is.



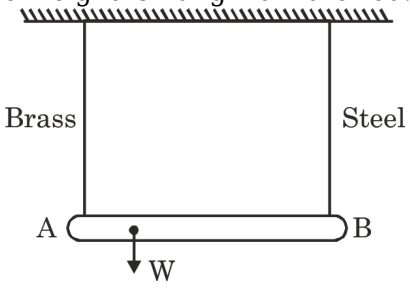
(A) $\rho = \frac{\rho_{min}}{\cos^3\alpha}$

(B) $\frac{\rho_{min}}{\cos^2\alpha}$

(C) $\frac{\rho_{min}}{\cos\alpha}$

(D) None of these

8) A 2 m long light metal rod AB is suspended from the ceiling horizontally by means of two vertical wires of equal length, tied to its ends. One wire is of brass and has cross-section of 0.3×10^{-4} m^2 and the other is of steel with 0.1×10^{-4} m^2 cross-section. In order to have equal stresses in the two wires, a weight is hung from the rod. The position of the weight along the rod from end A should be :



(A) 66.6 cm

(B) 133 cm

(C) 50 cm

(D) 155.6 cm

9) The radius r and height h of a cylinder are related to volume of the cylinder as $V= \pi r^2h$. If at an instant when radius is constant at 10 cm and height is increasing at a rate of 5 cm/s then find the rate of increase of volume of cylinder

(A) 100 π cm^3/s

(B) 1000π cm^3/s

(C) 50π cm^3/s

(D) 500π cm^3/s

10) A particle strikes a smooth horizontal surface at an angle of 45° with a velocity of 100 m/s and rebounds. If the coefficient of restitution between the surface and the particle is $\frac{1}{\sqrt{3}}$ then the angle, which the velocity of the particle after it rebounds will make with the surface, is

(A) 30°

(B) 45°

(C) 60°

(D) 90°

SECTION-I (ii)

1) A set of n-identical cubical blocks lies at rest along a line on a smooth horizontal surface. The separation between any two adjacent blocks is L. The block at one end is given a speed V towards the next one at time t = 0. All collisions are completely inelastic, then :-

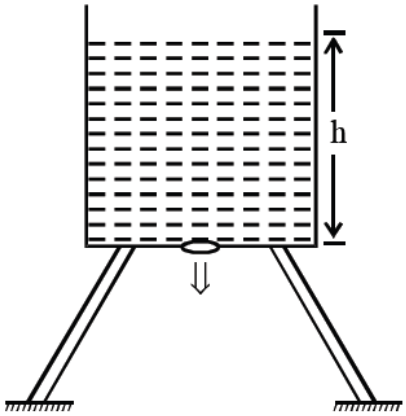
(A) The last block starts moving at $t = \frac{L}{2V}$

(B) The last block starts moving at $t = \frac{L}{V}$

(C) The centre of mass of the system will have a final speed V/n

(D) The centre of mass of the system will have a final speed V

2) A small hole of area a is at the bottom of a container of area A. The liquid is filled up to height h from base. As liquid comes out then $\left[a = \frac{A}{3}\right]$.



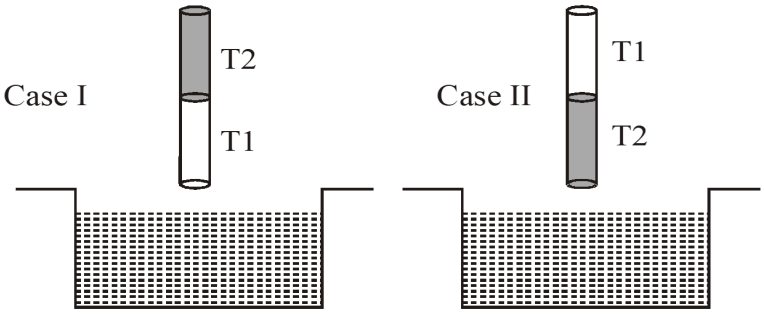
(A) level of liquid in container falls at rate of $\sqrt{\frac{gh}{4}}$ m/s

(B) magnitude of acceleration of top surface of liquid is $\frac{g}{8}$ m/s^2

(C) acceleration of top surface of liquid is $\frac{g}{3}$ m/s^2

(D) None of these

3) A cylindrical capillary tube of 0.2 *mm* radius is made by joining two capillaries T_1 and T_2 of different materials having water contact angles of 0° and 60°, respectively. The capillary tube is dipped vertically in water in two different configurations, case I and II as shown in figure. Which of the following option(*s*) is(are) correct ?
(Surface tension of water = 0.075 N/m, density of water = 1000 kg/m^3, take g = 10 m/s^2)



(A) The correction in the height of water column raised in the tube, due to weight of water contained in the meniscus, will be different for both cases.

(B) For case I, if the capillary joint is 5 *cm* above the water surface, the height of water column raised in the tube will be more than 8.75 *cm*. (Neglect the weight of the water in the meniscus)

(C) For case I, if the joint is kept at 8 *cm* above the water surface, the height of water column in the tube will be 7.5 *cm*. (Neglect the weight of the water in the meniscus)

(D) For case II, if the capillary joint is 5 *cm* above the water surface, the height of water column raised in the tube will be 3.75 *cm*. (Neglect the weight of the water in the meniscus)

4) A shell is fired vertically upward with a speed of 60 m/s. When it is at maximum height, it explodes into large number of fragments. Assume that the fragments fly in every possible direction and all of them have same initial speed of 25 m/s.
(1) The speed v (in m/s) of the center of mass of the system after 1 sec of explosion will be:-
(2) The minimum time t (in sec) taken by any one fragment to reach at ground will be :-

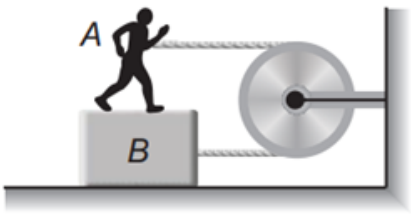
(A) v = 10 m/s

(B) v = 20 m/s

(C) t = 6s

(D) t = 4s

5) As shown in the figure, A is a man of mass 60 kg standing on a block B of mass 40 kg kept on ground. The coefficient of friction between the feet of the man and the block is 0.3 and that between B and the ground is 0.2. If the person pulls the string with 125 N force, then



(A) B will slide on ground

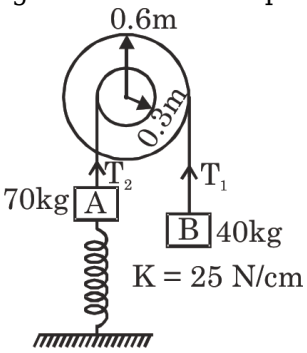
(B) A and B will move with acceleration 0.5 ms^{-2}

(C) the force of friction acting between A and B will be 125 N

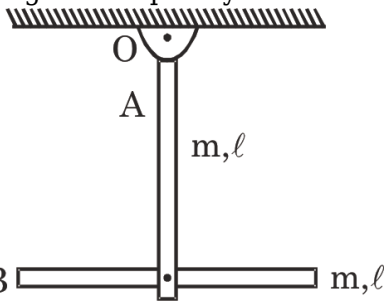
(D) the force of friction acting between B and ground will be 250 N

SECTION-III

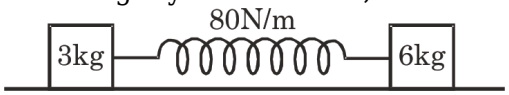
1) The figure shown is in equilibrium. Find out the extension in the spring (in cm). Pulley is fixed at centre.



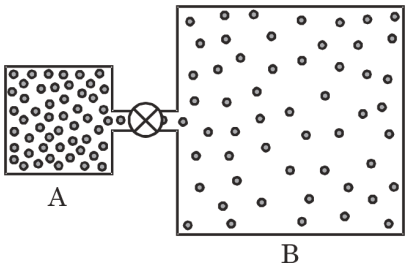
2) Two rods A and B (mass m and length ℓ each) are connected as shown. Rod A is free to rotate about O. Mid point of rod B is connected to lower end of rod A and free to rotate relative to A in the plane of paper. Angular frequency of oscillation of small angular amplitude of rod A is $\sqrt{\frac{Ng}{8\ell}}$. Find the value of N.



3) If the system shown in the figure is compressed slightly and released, then the time period (in sec) of oscillation will be : [Take : $\pi = \sqrt{10}$]

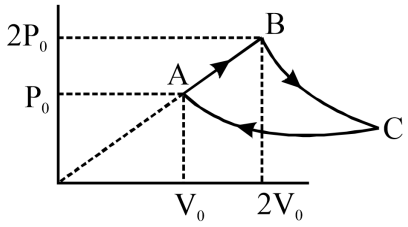


4) Container A in figure holds an ideal gas at a pressure of 1.0×10^5 Pa and a temperature of 300K. It is connected by thin tube (and a closed valve) to container B, with four times the volume of A. Container B holds the same ideal gas at a pressure of 9.0×10^5 Pa and a temperature of 400K. The valve is opened to allow the pressures to equalize, but the temperature of each container is maintained as before. What then is the common pressure in 10^5 Pa?



5) One mole of a diatomic gas is taken around a cyclic process shown in the figure. The process BC is adiabatic expansion and the process CA is isothermal compression. If the volume of the gas at C

0



is given by $V_C = (2)^n V$, then the value of n is

PART 2 : CHEMISTRY

SECTION-I (i)

1) The entropy change can be calculated by using the expression $\Delta S = \frac{q_{rev}}{T}$. When water freezes in a glass beaker, choose the correct statement amongst the following:

(A) Δs (system) decreases but Δs (surroundings) remains the same.
(B) Δs (system) increases but Δs (surroundings) decreases
(C) Δs (system) decreases but Δs (surroundings) increases
(D) Δs (system) decreases but Δs (surroundings) also decreases

2) Minimum mass of S_8 required to produce 320 gm of SO_3 with excess of O_2 gas from the following sequential reactions $S_8 + 8O_2 \xrightarrow{80\%} 8SO_2$ $2SO_2 + O_2 \xrightarrow{50\%} 2SO_3$

(A) 640 gm
(B) 160 gm
(C) 320 gm
(D) 200 gm

3) The number of litres of air required to burn 8 litre of C_2H_2 is approximately

(A) 40
(B) 60
(C) 80
(D) 100

4) Which of the following order of ionisation energy is correct ?

(A) $F_2 > F$
(B) $F_2 < F$
(C) $F_2 = F$
(D) Cannot predict

5) Comment on the C – C bond length for C_2H_6 and C_2F_6 compounds :

(A) $d_{C-C}(C_2H_6) > d_{C-C}(C_2F_6)$

(B) $d_{C-C}(C_2F_6) > d_{C-C}(C_2H_6)$

(C) $d_{C-C}(C_2F_6) = d_{C-C}(C_2H_6)$

(D) Can't be predicted

6) The photon emitted due to electronic transition from 5th excited state to 2nd excited state in Li^{2+}, is used to excite He^+ already in first excited state. He^+ ion after absorbing the photon reaches in an orbit having total energy equal to :

(A) $-3.4eV$

(B) $-13.6eV$

(C) $-6.8eV$

(D) $-27.2eV$

7) Consider the following nuclear reactions involving X & Y.

$X \rightarrow Y + {}_2^4He$

$Y \rightarrow {}_8O^{18} + {}_1H^1$

If both neutrons as well as protons in both the sides are conserved in nuclear reaction then moles of neutrons in 4.6 gm of X

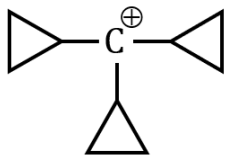
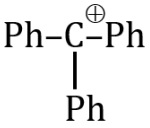
(A) 2.4 N_A

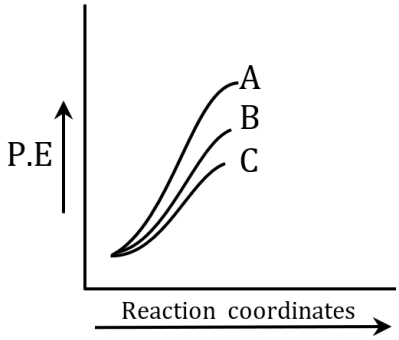
(B) 2.4

(C) 4.6

(D) 0.2 N_A

8) In the following graph, stability of different carbocations have been shown :

		
(I)	(II)	(III)



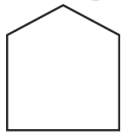
(A) I-A; II-B; III-C

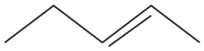
(B) I-B; II-C; III-A

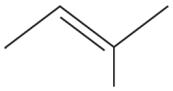
(C) I-C; II-B; III-A

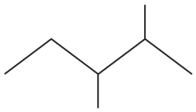
(D) I-B; II-A; III-C

9) Select pair of chain isomers from the following

(I) 

(II) 

(III) 

(IV) 

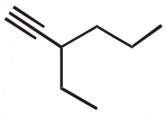
(A) I and II

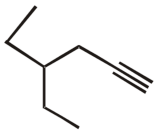
(B) II and IV

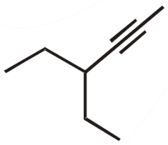
(C) I and IV

(D) II and III

10) Which alkyne will give 3-ethylhexane on catalytic hydrogenation ?

(A) 

(B) 

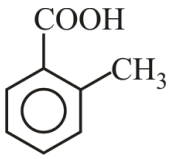
(C) 

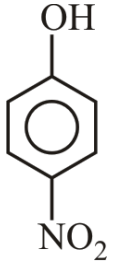
(D) All of these

SECTION-I (ii)

1) Which compound(s) is/are soluble in aqueous NaOH :

(A) 

(B) 

(C) 

(D) 

2) When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy, T_A (expressed in eV) and de–Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.20 V is $T_B = T_A - 1.50$ eV. If the de–Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$ then :

(A) the work function of A is 2.25 eV

(B) the work function of B is 3.70 eV

(C) $T_A = 2.00$ eV

(D) $T_B = 2.75$ eV

3) A sodium street light gives off yellow light that has a wavelength of 600 nm. Then :

(A) frequency of this light is 7×10^{14} s^{-1}.

(B) frequency of this light is 5×10^{14} s^{-1}.

(C) wave number of the light is 3×10^{6} m^{-1}.

(D) energy of the photon is approximately 2.07 eV.

4) Which of the following species is/are linear ?

(A) C_2H_2

(B) N_3^-

(C) SCN^-

(D) NO_2^+

5) If 27 g of Carbon is mixed with 88 g of Oxygen and is allowed to burn to produce CO_2 , then :

(A) Oxygen is the limiting reagent

(B) Volume of CO_2 gas produced at NTP is 50.4 L

(C) C and O combine in mass ratio 3 : 8.

(D) Volume of unreacted O_2 at STP is 11.2 L.

SECTION-III

1) For the reversible reaction $A(g) + B(s) \rightleftharpoons 2C(g)$, $\frac{K_p}{K_c} = (RT)^x$. Hence x is :

2) For reaction $XeF_6 + H_2O \rightleftharpoons XeOF_4 + 2HF$; $K_1 = 4$

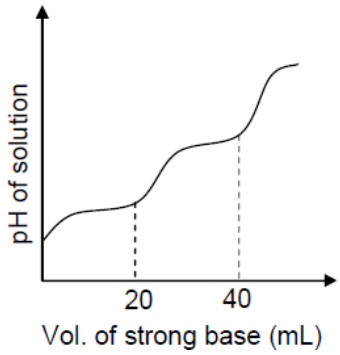
$XeO_4 + XeF_6 \rightleftharpoons XeOF_4 + XeO_3F_2$; $K_2 = 100$

Find equilibrium constant of

$$\frac{1}{2}XeO_4 + HF \rightleftharpoons \frac{1}{2}XeO_3F_2 + \frac{1}{2}H_2O$$

3) Pure gas 'A' was taken in a closed container at an initial pressure of 5 atm. The pressure in the container changed due to reactions $A(g) \overset{K_{p_1}}{\rightleftharpoons} B(g) + 2C(g)$ $C(g) \overset{K_{p_2}}{\rightleftharpoons} B(g) + D(g)$ If the total pressure due to mixture of gas A, B, C and D was found to be 10 atm, in which pressure due to C was 3 atm. Find K_P (atm^2) for first equilibria.
Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

4) 10 mL of H_2A (weak diprotic acid) solution is titrated against 0.1 M NaOH. pH of the solution is plotted against volume of strong base added and following observation is made.



If pH of the solution at first equivalence point is pH_1 and at second equivalence point is pH_2.
Calculate the value of $\frac{pH_2}{pH_1}$ at 25°C
Given : For H_2A, $pK_{a_1} = 4.6$ and $pK_{a_2} = 8$, log 25 = 1.4
(Answer upto nearest integer)

5) Find the total number of species where octet rule is not applicable.
BrF_5, SF_6, IF_7, CO, $BeCl_2$, BF_3, N_2O

PART 3 : MATHEMATICS

SECTION-I (i)

1) The equation of the circle which passes through the intersection of circles $x^2 + y^2 + 13x - 3y = 0$ and $2x^2 + 2y^2 + 4x - 7y - 25 = 0$ and whose centre lies on $13x + 30y = 0$, is -

(A) $x^2 + y^2 + 30x - 13y - 25 = 0$
(B) $4x^2 + 4y^2 + 30x - 13y - 25 = 0$
(C) $2x^2 + 2y^2 + 30x - 13y - 25 = 0$
(D) $x^2 + y^2 + 30x - 13y + 25 = 0$

2) Locus of mid points of the chords of contact of the pair of tangents to the circle $x^2 + y^2 = 2$ drawn from the points on the line $3x + 4y = 10$ is :-

(A) $x^2 + y^2 - 3x - 4y = 0$

(B) $5x^2 + 5y^2 + 3x + 4y = 0$

(C) $x^2 + y^2 - 3x + 4y = 0$

(D) $5x^2 + 5y^2 - 3x - 4y = 0$

3) For any three positive real numbers a, b and c, $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then :

(A) a, b and c are in G.P.

(B) b, c and a are in G.P.

(C) b, c and a are in A.P.

(D) a, b and c are in A.P.

4) If α, β & γ are the roots of the equation $x^3 - x - 1 = 0$ then, $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma}$ has the value equal to:

(A) zero

(B) -1

(C) -7

(D) 1

5) If p, q, r, s ∈ R, then equation $(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ has

(A) 6 real roots

(B) at least two real roots

(C) 2 real and 4 imaginary roots

(D) 4 real and 2 imaginary roots

6) If the coefficients of x^{-2} and x^{-4} in the expansion of $\left(x^{\frac{1}{3}} + \frac{1}{2x^{\frac{1}{3}}}\right)^{18}$, (x > 0) are m and n respectively, then $\frac{m}{n}$ is equal to :

(A) $\frac{4}{5}$

(B) $\frac{5}{4}$

(C) 182

(D) 27

7) 5 Indian & 5 American couples meet at a party & shake hands . If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hand shakes that takes place in the party is :

(A) 95

(B) 110

(C) 135

(D) 150

8) If $x\cos\alpha + y\sin\alpha = P$, be a variable chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{2a^2} = 1$ which subtends a right angle at the centre of the hyperbola, then the chords touch a fixed circle with centre origin and radius r, then r =

(A) a

(B) $\sqrt{2}a$

(C) 2a

(D) 4a

9)

Number of possible values of (x,y) when $\sin^4 x + 16\cos^4 y + 2 = 8\sin x \cos y$, $x, y \in (0,\pi)$

(A) 1

(B) 2

(C) 3

(D) 4

10) If $\sin\theta + \sin^2\theta + \sin^3\theta = 2$ then the value of $\cos^6\theta - 4\cos^4\theta + 10\cos^2\theta$ is

(A) 4

(B) 3

(C) 2

(D) 1

SECTION-I (ii)

1) If H(3,4) and O'(1,2) are respectively be the orthocenter and circumcenter of ΔABC and equation of side AB is x – y + 7 = 0, then

(A) centroid of ΔABC is $\left(\frac{5}{3}, \frac{8}{3}\right)$

(B) equation of circumcircle of ΔABC is $(x-1)^2 + (y-2)^2 = 80$

(C) equation of circumcircle of ΔABC is $(x-1)^2 + (y-2)^2 = 100$

(D) there will be a unique ΔABC

2) A(3,4) B(0, 0) and C(3, 0) are vertices of ΔABC. If 'P is a point inside ΔABC such that d(P, BC) ⩽ min. {d(P, AB), d(P, AC)}, where d(P, BC) represents distance of P from BC, then identify correct statement-

(A) if d(P, BC) is maximum, then P is incentre of the triangle.

(B) if d(P, BC) is maximum, then P is centroid of the triangle.

(C) max. d(P, BC) = 1

(D) max. d(P, BC) = $\frac{4}{3}$

3)

Let for n ∈ **N** ; $(2x^2 - 2x + 1)^n = \sum_{r=0}^{2n} (-1)^r a_r x^r$ then

(A) $\sum_{r=0}^{2n} a_r = 1$

(B) $\sum_{r=1}^{2n} (-1)^r a_r = 0$

(C) $\sum_{k\geq 0} {}^nC_k \, {}^{2n-2k}C_r = 2^{n-r} a_r$ (wherever defined)

(D) $\sum_{r=0}^{2n} \left((-1)^r a_r a_{2n-r}\right) = 0$ if n = odd

4) Which of the following is are true?

(A) The value of ${}^{404}C_4 - {}^4C_1 \cdot {}^{303}C_4 + {}^4C_2 \cdot {}^{202}C_4$ is equal to $(404)^4$

(B) Let $(1 + x + x^2)^{25} = a_0 + a_1x + a_2x^2 \ldots\ldots a_{50} \cdot x^{50}$, then the value of $a_3 - a_5 + a_7 \ldots\ldots\ldots - a_{49}$ is 24

(C) Let $(1 + x + x^2)^{25} = a_0 + a_1x + a_2x^2 \ldots\ldots a_{50} \cdot x^{50}$, then the value of $a_0 - a_2 + a_4 \ldots\ldots - a_{50} = 0$

(D) The constant term in the expansion of $\left(x + x^2 + \frac{1}{x} + \frac{1}{x^2}\right)^{15}$ is $\sum_{r=0}^{5} \left({}^{15}C_{5+r}\right)\left({}^{15}C_{3r}\right)$

5) If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$ then

(A) $x_1 + x_2 + x_3 + x_4 = 0$

(B) $y_1 + y_2 + y_3 + y_4 = 0$

(C) $x_1x_2x_3x_4 = c^4$

(D) $y_1\, y_2\, y_3\, y_4 = c^4$

SECTION-III

1) If n is the number of ways in which an examiner can assign 30 marks to 8 questions, giving not less than 2 marks to any question such that $m = \frac{n}{{}^{20}C_6}$, then the value of m is

2) Radius of largest circle that can be inscribed in the parabola $x^2 = 4y$ and passing through its focus is

3) The area of the trapezium whose vertices lie on the parabola $y^2 = 4x$ and its diagonals pass through (1, 0) and having length $\frac{25}{4}$ units each is equal to $\frac{25k}{4}$ then k is

4) If α and β are angles such that $\alpha + \beta = \frac{2\pi}{3}$, then number of integers in range of (sin α – sin β + 4 cosα cosβ) is

5) If $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a+b+c}$, then the value of $\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)^{2017} - \frac{2017}{a^{2017} + b^{2017} + c^{2017}} + \frac{2016}{(a+b+c)^{2017}}$ is

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4	5	6	7	8	9	10
A.	B	A	B	B	C	C	A	C	D	A

SECTION-I (ii)

Q.	11	12	13	14	15
A.	A,C	A,B	A,C,D	A,D	A,B

SECTION-III

Q.	16	17	18	19	20
A.	4	9	1	7	6

PART 2 : CHEMISTRY

SECTION-I (i)

Q.	21	22	23	24	25	26	27	28	29	30
A.	C	C	D	B	A	A	B	C	D	D

SECTION-I (ii)

Q.	31	32	33	34	35
A.	A,B,C,D	A,B,C	B,D	A,B,C,D	B,C,D

SECTION-III

Q.	36	37	38	39	40
A.	1	5	9	2	5

PART 3 : MATHEMATICS

SECTION-I (i)

Q.	41	42	43	44	45	46	47	48	49	50
A.	B	D	C	C	B	C	C	B	A	B

SECTION-I (ii)

Q.	51	52	53	54	55
A.	A,B,D	A,C	B,C,D	B,C,D	A,B,C,D

SECTION-III

Q.	56	57	58	59	60
A.	**3**	**4**	**3**	**6**	**0**

SOLUTIONS

PART 1 : PHYSICS

1)

$Fx - \frac{1}{2}kx^2 - \mu_1 m_1 gx = 0$

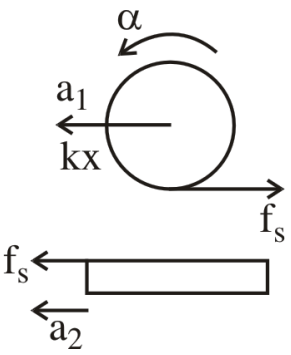
$Kx = \mu_2 m_2 g$

2)

W.E.T

$\frac{1}{2}ka^2 - \mu mga = 0$

$k = \frac{2\mu mg}{a}$



3)

$kx - f_s = ma_1$ .....(1)

$f_s R = \frac{mR^2}{2}\alpha$ .....(2)

$f_s = ma_2$ .....(3)

$a_1 - \alpha r = a_2$ .....(4)

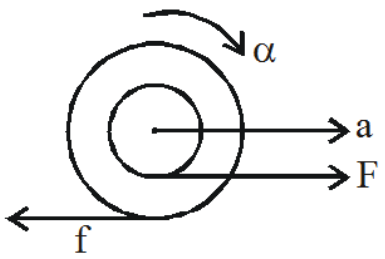
from (2) & (3)

$ma_2 R = \frac{mR^2}{2}\alpha$

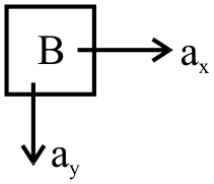
$a_2 = \frac{\alpha R}{2}$

from equation (4)

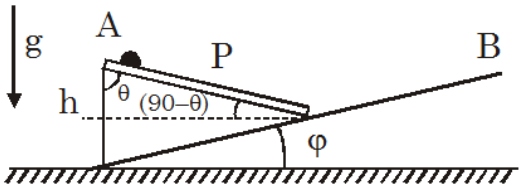
$a_1 - 2a_2 = a_2 \Rightarrow a_1 = 3a_2$



4)

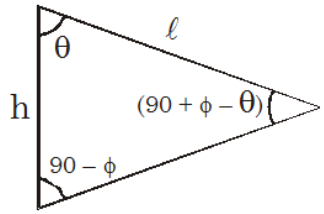


5) Vertical component of acceleration of B will be equal to acceleration of A



6)

$a_{ball} = g\sin(90 - \theta) = g\cos\theta$



$$\frac{\ell}{\sin(90-\phi)} = \frac{h}{\sin(90+\phi-\theta)}$$

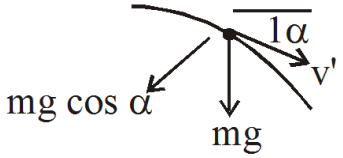
$$\ell = \frac{h\cos\phi}{\cos(\phi-\theta)}$$

$$t = \sqrt{\frac{2\ell}{g\cos\theta}} = \sqrt{\frac{2h\cos 60^\circ}{g\cos\theta\cos(60-\theta)}}$$

t_{min} for $\theta = 30°$

7) **Case - I** → At highest pt.

$$mg = \frac{m.(v\cos\theta)^2}{\rho_{min}} \Rightarrow \rho_{min} = \frac{v^2\cos^2\theta}{g}$$



Case - II At given pt.

Equating velocity in x = direction ⇒ v cosθ = v'cosα

$$\Rightarrow v' = \frac{v\cos\theta}{\cos\alpha}$$

$$\Rightarrow mg\cos\alpha = \frac{m.(v^1)^2}{\rho}$$

$$\Rightarrow \rho = \frac{v^2\cos^2\theta}{g.\cos^3\alpha} \Rightarrow \rho = \frac{\rho_{min}}{\cos^3\alpha}$$

8) $\sigma = \frac{F}{A}$

so $F_{brass} = 3\ F_{steel}$

so $F_{brass}\ x = F_{steel}\ (2 - x) \Rightarrow x = 0.5$ m

9) $\frac{dv}{dt} = \pi r^2 \left(\frac{dh}{dt}\right)$

$= (5) \times 100\pi$

$= 500\pi$

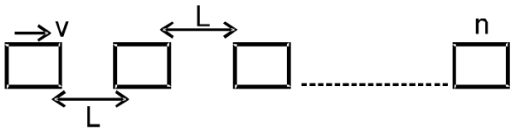
10)

$100 \cos 45° = V_x$ ....(1)

$100 \sin 45° = e(V_y)$ ....(2)

$\tan \theta = \frac{V_x}{V_y}$ ....(3)

11)



$mv = nv' m \Rightarrow v' = \frac{v}{n}$

time for first collision is $t_1 = \frac{L}{V}$

2nd collisions $t_2 = \frac{2\ell}{V} = 2t_1$

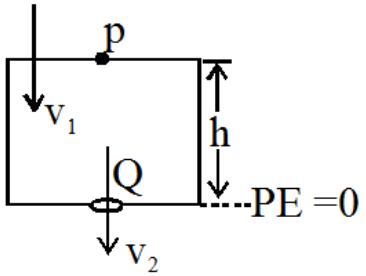
so $t = t_1 + 2t_1 + 3t_1 + at_1$ .....(n–1) t_1.

$t = t_1 [1 + 2 + 3]$ .....(n–1)]

$= \frac{(n-1)(n-1+1)}{2} = \frac{n(n-1)}{2}$

so $t = \frac{L}{2V} n\ (n - 1)$

12) Applying Bernoulli's equation between P and Q



$p_0 + \rho gh + \frac{1}{2}\rho v_1^2 = \rho_0 + 0 + \frac{1}{2}\rho v_2^2$

$\rho gh = \frac{\rho}{2}\left(v_2^2 - v_1^2\right)$

$v_2^2 - v_1^2 = 2gh$ .....(i)

From equation of continuity,

$Av_1 = av_2 = \frac{A}{3}v_2$

$\Rightarrow v_2 = 3v_1$

$9v_1^2 - v_1^2 = 2gh$

$8v_1^2 = 2gh$

$\Rightarrow v_1 = \sqrt{\frac{gh}{4}}$

Acceleration of top layer

$a = \frac{dv_1}{dt} = \frac{1}{2}\sqrt{g} \times \frac{1}{2\sqrt{h}} \times \frac{dh}{dt}$

$= \frac{1}{4}\sqrt{\frac{g}{h}} \times (--v_1) = -\frac{1}{4}\sqrt{\frac{g}{h}} \times \frac{\sqrt{gh}}{2} = -\frac{g}{8} m/s^2$

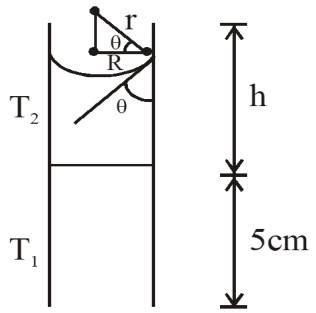
13) $h = \frac{2T\cos\theta}{\rho g R}$; $h_1 = \frac{2 \times 0.075 \times \cos 0^\circ}{1000 \times 10 \times 0.2 \times 10^{-3}}$

$\Rightarrow h_1 = 75\ mm$ (in T_1) [If we assume entire tube of T_1]

$\Rightarrow h_2 = \frac{2 \times 0.075 \times \cos 60^\circ}{1000 \times 10 \times 0.2 \times 10^{-3}} = 37.5 mm (in T_2)$ [If we assume entire tube of T_2]

Option (1): Since contact angles are different so correction in the height of water column raised in the tube will be different in both the cases, so option (1) is correct

Option (2): If joint is 5 *cm* is above water surface, then let's say water crosses the joint by height h, then:



$\Rightarrow P_0 - \frac{2T}{r} + \rho g h + \rho g \times 5 \times 10^{-2} = P^0$

$\Rightarrow \cos\theta = \frac{R}{r}, r = \frac{R}{\cos\theta}$

$\Rightarrow \rho g(h + 5 \times 10^{-2}) = \frac{2T\cos\theta}{R}$

$\Rightarrow h = \frac{2 \times 0.075 \times \cos 60}{0.2 \times 10^{-3} \times 1000 \times 10} - 5 \times 10^{-2}$

$\Rightarrow h = -ve$, not possible, so liquid will not cross the interface, but angle of contact at the interface will change, to balance the pressure,

So, option (2) is wrong.

Option (3) : If interface is 8 *cm* above water then water will not even reach the interface, and water will rise till 7.5 *cm* only in T_1, so option (3) is right.

Option (4) : If interface is 5 *cm* above the water in vessel, then water in capillary will not even reach the interface. Water will reach only till 3.75 *cm*, so option (4) is right.

14)

Maximum height $= \frac{u^2}{2g} = \frac{60^2}{20} = 180 m$

At the time of explosion velocity of centre of mass will be zero.

So the speed after 1 sec equals to 10 m/s down wards.

for minimum time : Any one fragment which has initial velocity 25 m/s downwards will take minimum time to reach at ground, so

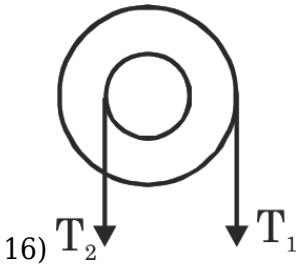
$\therefore \quad -180 = -25t - \frac{1}{2}gt^2$

15)

Force on A and B = 250 N – f_{BG}

= 50 N

$\Rightarrow a_{AB} = 0.5\ m/s^2$



16)

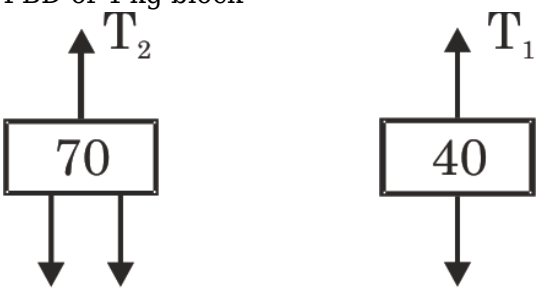
FBD of pulley

$T_2(0.3) = T_1(0.6)$

$T_2 = 2T_1$ .....(1)

FBD of 70 kg block

FBD of 4 kg block



$T_2 = 70g + kx$ $T_1 = 40g$

17) $\tau = (mg\frac{\ell}{2} + mg\square)\sin\theta = \frac{3mg\ell}{2}\sin\theta$

$I = \frac{m\ell^2}{3} + m\ell^2 = \frac{4m\ell^2}{3}$

$\therefore \alpha = \frac{3mg\ell \times 3\sin\theta}{2 \times 4m\ell^2} = \frac{9}{8}\frac{g}{\ell}\sin\theta$

$\therefore \omega = \sqrt{\frac{9g}{8\ell}}$ (for small oscillation $\sin\theta \approx \theta$)

18) $T = 2\pi\sqrt{\frac{\mu}{k}}$

$u = \frac{3 \times 6}{3+6} = 2$

$$T = 2\pi\sqrt{\frac{2}{80}}$$

$\Rightarrow$ 1 sec

19) $$\frac{P_1V_1}{T_1} + \frac{P_2V_2}{T_2} = \frac{PV_1}{T_1} + \frac{PV_2}{T_2}$$

Putting values & solving, we get :-

$P = 7 \times 10^5$ Pa

20) $(2P_0)(2V_0)^{\gamma} = P_C V_C^{\gamma}$ .....(i)

$P_0V_0 = P_CV_C$ .....(ii)

For diatomic gas $\gamma = \frac{7}{5}$

$\therefore V_C = \left[(2)^6\right] V_0$

PART 2 : CHEMISTRY

21) ΔS (system) decreases but ΔS (Surrounding increases)

22) Mass of S_8 Req. = $\frac{320}{80}\left(\frac{2}{2}\right) \times \frac{1}{0.5} \times \frac{1}{8} \times \frac{1}{0.8} \times (32 \times 8)$ = 320 gm

23) $$C_2H_2 \overset{+5/O_2}{\underset{2}{}} \rightarrow 2CO_2 + H_2O$$

1 litre C_2H_2 required 5/2 O_2 litre

8 litre require

$= 8 \times \frac{5}{2} = \frac{40}{2} = 20$ litre O_2

1 litre air contain 0.2 litre O_2

Volume of air contain 20 litre

$= \frac{20 \text{ litre}}{0.2} = 100$ litre air

24) Last $\mathbf{e}^{\circ}$ in $\mathbf{F_2}$ is being removed from ABMO.

25)

On the basis on bent's rule

26) For Li^{2+}, n = 6 to n = 3

For H, the similar transition is 2 to 1

Fore He^{+}, the similar transition is 4 to 2

Energy of 4th orbit of He^+ = $\frac{-13.6 \times 2^2}{4^2} = -3.4eV$

27) A = no. of proton + no. of neutron

$^{A}_{Z}X$

Z = no. of proton = no. of electron

So Y = $_8O^{18}$ + $_1H^1$

n_{proton} in 'Y' = 8+1 = 9

$n_{proton} + n_{neutron} = 19$

$n_{neutron} = 10$

So it will be $_9Y^{19}$

So for X = $^{19}_{9}Y$ + $_2He^4$

It will be $^{23}_{11}X$

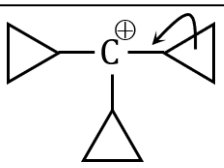
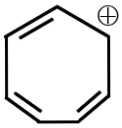
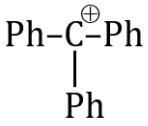
So no. of neutron = 23–11 = 12 in 1atom

Given wt = 4.6gm Atomic wt. = A = 23

$n_{moles} = \frac{4.6}{23} = 0.2$

So total neutron = $0.2 \times 12 \times N_A = (2.4N_A)$ so (A) is correct

28)

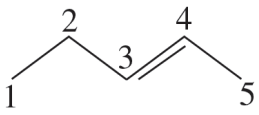
		
(I)	(II)	(III)
σ-bond resonance	Aromatic C^+	Resonance stabilised
Dancing Resonance		least stable

⇒ stability α $\frac{1}{P.E}$

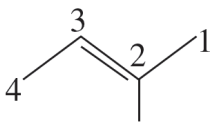
⇒ P.E. order : III > II > I

⇒ I :- C II :- B III :- A

29)



(II)



(III)

P.C differ so they becomes Chain isomer

30) All alkyne on catalytic hydrogenation give 3-ethylhexane.

31) Compound which is more acidic then H_2O is soluble in aq. NaOH.

32) $T_A = 4.25 - W_A$
$T_B = 4.20 - W_B$

$$\lambda_A = \sqrt{\frac{150}{T_A}}$$

$$\lambda_B = \sqrt{\frac{150}{T_B}}$$

$$2\lambda_A = \sqrt{\frac{150}{T_A - 1.50}}$$

$T_A = 2$ eV, $T_B = 0.5$ eV, $W_A = 2.25$ eV, $W_B = 3.7$ eV

33)

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{600 \times 10^{-9}} = 5 \times 10^{14} \text{ sec}^{-1}$$

$$E = \frac{12400}{6000} = 2.07 \text{ eV.}$$

34) (A) HC ≡ CH
(B) N ≡ N : → N^- or N^- = N^+ = N^-
(C) S = C = N^-
(D) O = N^+ = O

35)

	C	+	$O_2 \rightarrow CO_2$
mass	27		88
moles	$\frac{27}{12}$		$\frac{88}{32}$

C is limiting reagent

Moles of CO2 produced = moles of C = $\frac{27}{12} = 2.25$
 Volume of CO_2 at STP = 2.25 x 22.4 = 50.4 L
Ratio of C and O in CO_2 = 12 : 32 = 3 : 8
Moles of unreacted O2 = 2.75 – 2.25 = 0.5
 Volume of unreacted O_2 at STP = 0.5 × 22.4 = 11.2 L

36) $\Delta n_g = 1$So, $\frac{K_P}{K_C} = (RT)^1$

37) $\frac{1}{2}(XeO_4 \times XeF_6 \rightleftharpoons XeO_4 + XeO_3F_2)$ $K = (100)^{\frac{1}{2}}$

$\frac{1}{2}(XeOF_4 + 2HF \rightleftharpoons XeF_6 + H_2O)$ $K = \left(\frac{1}{4}\right)^{\frac{1}{2}}$

$\frac{1}{2}XeO_4 + HF \rightleftharpoons \frac{1}{2}XeO_3F_2 + \frac{1}{2}H_2O$ $K = \left(\frac{100}{4}\right)^{\frac{1}{2}} = 5$

38) $\underset{5-x}{A(g)} \rightleftharpoons \underset{x+y}{B(g)} + \underset{2x-y}{2C(g)}$

$\underset{2x-y}{C(g)} \rightleftharpoons \underset{x+y}{B(g)} + \underset{y}{D(g)}$

$2x - y = 3$ ....(i)

$5 - x + x + y + 2x - y + y = 12$

$5 + y + 2x = 10$ ....(ii)

$x = 2, y = 1$

$(K_p)_1 = \dfrac{(2x-y)^2(x+y)}{(5-x)} = 9$

39) $H_2CO_3 + NaOH \rightarrow NaHCO_3 + H_2O$

$pH_1 = \dfrac{1}{2}(pK_{a_1} + pK_{a_2}) = \dfrac{1}{2}(4.6 + 8) = 6.3$

$N_1V_1 = N_2V_2$

$N_1 \times 10 = 0.1 \times 20$

$N_1 = 0.2$

$NaHCO_3 + NaOH \rightarrow Na_2CO_3 + H_2O$

$c = \dfrac{2m\,mole}{50mL} = \dfrac{1}{25}M$

$\Rightarrow pH = \dfrac{[14 + 8 - 1.4]}{2} = 10.3$

$\dfrac{pH_2}{pH_1} = \dfrac{10.3}{6.3} = 1.63$

40) Octet rule is not applicable on hypovalent, hypervalent and odd electron species.
So, only applicable for CO, N_2O.

PART 3 : MATHEMATICS

41) Equation of circle passing through point of intersection of two given circles will be

$(x^2 + y^2 + 2x - 7y/2 - 25/2) + \lambda (x^2 + y^2 + 13x - 3y) = 0$

$\Rightarrow x^2(1 + \lambda) + y^2(1 + \lambda) + x(2 + 13\lambda) - y\left(\dfrac{7}{2} + 3\lambda\right) - \dfrac{25}{2} = 0$

centre $= \left(\dfrac{(2 + 13\lambda)}{-2(1 + \lambda)}, \dfrac{7/2 + 3\lambda}{2(1 + \lambda)}\right)$

 centre lies on $13x + 30y = 0$

$\Rightarrow -13\left(\dfrac{2 + 13\lambda}{2(1 + \lambda)}\right) + 30\left(\dfrac{7/2 + 3\lambda}{2(1 + \lambda)}\right) = 0$

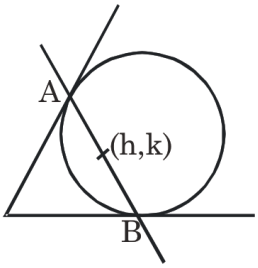
$\Rightarrow -26 - 169\lambda + 105 + 90\lambda = 0$

$= \lambda = 1$

 Required circle

$2x^2 + 2y^2 + 15x - \dfrac{13}{2}y - \dfrac{25}{2} = 0$

$\Rightarrow 4x^2 + 4y^2 + 30x - 13y - 25 = 0.$

42) 

Equation of AB is T = 0

$(10y - 8) + \lambda(4x - 3y) = 0$

It passes through $\left(\frac{3}{5}, \frac{4}{5}\right)$

Again equation of AB is $T = S_1$

$xh + yk = h^2 + k^2$

$x^2 + y^2 = \frac{3x}{5} + \frac{4y}{5}$

$5x^2 + 5y^2 - 3x - 4y = 0$

43) $(15a)^2 + (3b)^2 + (5c)^2 - (15a)(5c) - (15a)(3b) - (3b)(5c) = 0$

$\frac{1}{2}[(15a - 3b)^2 + (3b - 5c)^2 + (5c - 15a)^2] = 0$

it is possible when $15a = 3b = 5c$

$\Rightarrow b = \frac{5c}{3}, a = \frac{c}{3}$

$a + b = 2c$

$\Rightarrow$ b, c, a in A.P.

44)

$x^3 - x - 1 = 0 \Rightarrow \alpha, \beta, \gamma$ then $a^3 - a - 1 = 0$ ...(1)

Let $\frac{1+\alpha}{1-\alpha} = y \Rightarrow \alpha = \frac{y-1}{y+1}$ from equation (1) $\left(\frac{y-1}{y+1}\right)^3 - \left(\frac{y-1}{y+1}\right) - 1 = 0$

$\Rightarrow \frac{1+\alpha}{1-\alpha}, \frac{1+\beta}{1-\beta}, \frac{1+\gamma}{1-\gamma}$

$y^3 + 7y^2 - y + 1 = 0$

then $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma} = -7$

45)

Dis. of $x^2 + px + 3q$ is $p^2 - 12q \not< D1$

Dis. of $-x^2 + rx + q$ is $r^2 + 4q$ º D2
Dis. of $-x^2 + sx - 2q$ is $s^2 - 8q$ º D3
Case 1 : If q < 0, then D1 > 0, D3 > 0 and D2 may or may not be positive
Case 2 : If q > 0, then D2 > 0 and D1, D3 may or may not be positive
Case 3 : If q = 0, then D1 ³ 0, D2 ³ 0 and D3 ³ 0
from **Case 1, Case 2** and **Case 3** we can say that the given equation has at least two real roots.

46)

$$T_{r+1} = {}^{18}C_r \left(x^{\frac{1}{3}}\right)^{18-r} \left(\frac{1}{2x^{\frac{1}{3}}}\right)^r$$

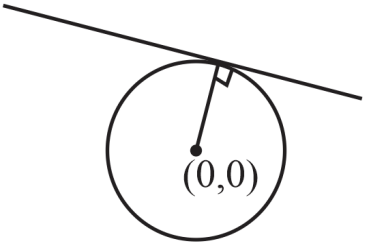
$$x^{\frac{18-2r}{3}} \cdot \frac{1}{2^r} \cdot {}^{18}C_r$$

$$\frac{18-2r}{3} = -2, -4 \qquad r = 12, 15$$

$$\frac{m}{n} = \frac{2^{-12}\,{}^{18}C_{12}}{2^{-15}\,{}^{18}C_{15}} = 182$$

47) Both men = $^{10}C_2 = 45$
All men & American women = 5 × 9 = 45
Indian women & American women = $^{10}C_2 = 45$
Total = 135

48) Using concept of homogenising,



$$\frac{x^2}{a^2} - \frac{y^2}{2a^2} - 1^2 = 0$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{2a^2} - \left(\frac{x\cos\alpha + y\sin\alpha}{P}\right)^2 = 0$$

$\theta = 90^\circ$
$\Rightarrow$ coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow \frac{1}{a^2} - \frac{\cos^2\alpha}{P^2} - \frac{1}{2a^2} - \frac{\sin^2\alpha}{P^2} = 0$$

$$\Rightarrow \frac{1}{2a^2} = \frac{1}{P^2} \Rightarrow P = \sqrt{2}a$$

$\Rightarrow$ radius $= \sqrt{2}a$

49)

Apply AM ≥ GM
For $\sin^4 x$, $16\cos^4 y$, 1, 1

$$\frac{\sin^4x + 16\cos^4y + 1 + 1}{4} \geq \sqrt[4]{\sin^4x.16\cos^4y}$$

$\Rightarrow \sin^4x + 16\cos^4y + 2 \geq 8\,|\sin x|\,|\cos y|$ ...(1)

$x \in (0, \pi)$ So $\sin x > 0$

& $\cos y > 0$ (L.H.S. > 0)

$\Rightarrow y \in \left(0, \frac{\pi}{2}\right)$ .... (2)

From (1) & (2)

$\sin^4x + 16\cos^4y + 2 = 8\sin x\cos y$ (AM = GM)

possible when $\sin^4x = 1 = 16\cos^4y = 1$

$\Rightarrow \sin x = 1,\ \cos y = \frac{1}{2}$

$\Rightarrow x = \frac{\pi}{2}, y = \frac{\pi}{3}$

(x, y) $\in \left(\frac{\pi}{2}, \frac{\pi}{3}\right)$

50) $\sin\theta + \sin^2\theta + \sin^3\theta = 2$

$\sin\theta(1 + \sin^2\theta) = 2 - \sin^2\theta = \cos^2\theta + 1$

Squaring both sides

$\Rightarrow \sin^2\theta(1 + \sin^2\theta)^2 = \cos^4\theta + 2\cos^2 + 1$

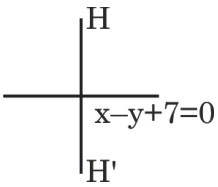
$\Rightarrow (1 - \cos^2\theta).(1 + 1 - \cos^2\theta)^2 = \cos^4\theta + 2\cos^2x + 1$

$\Rightarrow (1 - \cos^2\theta).(4 + \cos^4\theta - 4\cos^2\theta) = \cos^4\theta + 2\cos^2x + 1$

$\Rightarrow \cos^6\theta - 4\cos^4\theta + 10\cos^2\theta = 3$

51) Image of orthocenter about base lies on circumcircle

$$\frac{x-3}{1} = \frac{y-4}{-1} = -2\left(\frac{3-4+7}{1+1}\right)$$

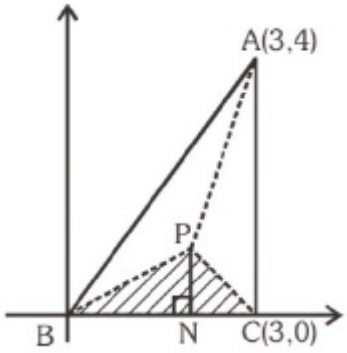


x = –3 , y = 10

H'(–3, 10)

$R = OH' = \sqrt{(1+3)^2 + (10-2)^2}$

$= \sqrt{64 + 16} = \sqrt{80}$



52)

$d(P, BC) \leq \min. \{d(P, AB), d(P, AC)\}$

The region represented is shown by shaded region.

where BP and CP are angle bisectors

Clearly maximum of d(P, BC) occurs, when P is incentre of ΔABC

$\therefore$ Maximum of d(P, BC) = PN = ordinate of incentre P= 1

53) Replace x by –x

$$\sum_{r=0}^{2n} a_r x^r$$

we get ; $(2x^2 + 2x + 1)^n =$

$$\sum_{r=0}^{2n} a_r = 5^n, \sum_{r=0}^{2n} (-1)^r a_r = 1$$

So,

Also, $a_0 = 1$

again,

$$\sum_{r=0}^{2n} a_r x^r = \left((x+1)^2 + x^2\right)^n = \sum {}^nc_k x^{2k}(x+1)^{2n-2k}$$

$$= \sum \left[{}^nc_k \sum {}^{2n-2k}c_t x^{2n-t}\right]$$

$\Rightarrow$ comparing coeff. of x^{2n-r}, we get

$$\sum {}^nc_k {}^{2n-2k}C_r = a_{2n-r} = 2^{n-r}a_r$$

Also,

$$\sum_{r=0}^{2n} (-1)^r . a_r a_{2n-r} = \text{coeff. of } x^{2n}$$

$$\text{in } \left((2x^2 - 2x + 1)^n (2x^2 + 2x + 1)^n\right)$$

= coefficient of x^{2n} in $(4x^4 + 1)^n = 0$ if 'n' is odd.

54) (A)

Coefficient of x^4 in $\left[{}^4C_0(1+x)^{404} - {}^4C_1(1+x)^{303}.....\right]$

= coefficient of x^4 in $((1 + x)^{101} - 1)^4$

$$= \left({}^{101}C_1 x + {}^{101}C_2 x^2 ..... {}^{101}C_{101} \cdot x^{101}\right)^4 = (101)^4$$

(B) $\left(1 + x + x^2\right)^{25} = a_0 + a_1 x + a_2 x^2 + a_3 x^3 ..... a_{50} \cdot x^{50}$ ....(i)

Put x = i both sides

$0 + i = (a_0 - a_2 + a_4 ..... - a_{50}) + i(a_1 - a_3 .....+ a_{49})$

compare real and imaginary part both sides

$a_0 - a_2 + a_4 ..... = 0$

$a_1 - a_3 + a_5 ..... + a_{49} = 1$

$a_3 - a_5 + a_7 ..... - a_{49} = a_1 - 1$ .... (ii)

Now for finding a_1 differentiate (1) then put x = 0

$25(1 + x + x^2)(1 + 2x)$

$= a_1 + 2a_2x + 3a_3x^2 ..... 50.a_{50}\cdot x^{49}$

x = 0

$25 = a_1$

from (ii)

$a_3 - a_5 ..... -a_{49} = 25 - 1 = 24$

(C)

$a_0 - a_2 + a_4 ..... -a_{50} = 0$

(D) $\left(x + x^2 + \frac{1}{x} + \frac{1}{x^2}\right)^{15}$

$= \left(x + x^2\right)^{15}\left(1 + \frac{1}{x^3}\right)^{15}$

$= x^{15}\left({}^{15}C_0 + {}^{15}C_1 x .... {}^{15}C_{15}\cdot x^{15}\right)\left({}^{15}C_0 + {}^{15}C_1\cdot\frac{1}{x^3} .... {}^{15}C_{15}\cdot\frac{1}{x^{45}}\right)$

Here constant term is

${}^{15}C_0\cdot{}^{15}C_5 + {}^{15}C_6\cdot{}^{15}C_3 + {}^{15}C_6\cdot{}^{15}C_7 + {}^{15}C_8\cdot{}^{15}C_9 + {}^{15}C_9\cdot{}^{15}C_{12} + {}^{15}C_{10}\cdot{}^{15}C_{15}$

$= \sum_{r=0}^{5}\left({}^{15}C_{r+5}\right)\left({}^{15}C_{3r}\right)$

55) Solving $xy = c^2$ and $x^2 + y^2 = a^2$, we have

$x^2 + \frac{c^4}{x^2} = a^2$

$\Rightarrow x^4 - a^2x^2 + c^4 = 0$

$\Rightarrow \sum x_i = 0$ and $x_1\, x_2\, x_3\, x_4 = c^4$

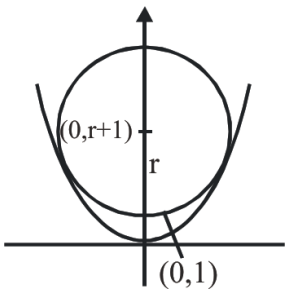
Similarly, if we climinate y, then $y_1\, y_2\, y_3\, y_4 = c^4$

56) The maximum marks that can be assigned to any question is 16. If x_i are the marks assigned to *i*th question, then $x_1 + x_2 + ... + x_8 = 30$ and $2 \le x_i \le 16$ for i = 1, 2, ..., 8.

Put $x_i = y_i + 2$, so that $y_i \ge 0$ and $y_1 + y_2 + ... + y_8 = 14$

The number of non-negative integral solutions of this equations is $m = {}^{14+8-1}C_{8-1} = {}^{21}C_7 \Rightarrow m = 3$

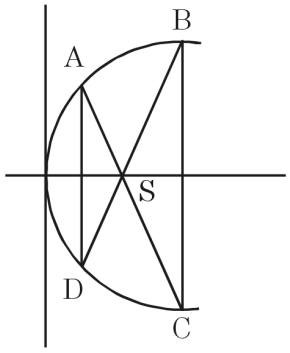
57)



Equation of circle : $x^2 + (y - r - 1)^2 = r^2$

solve with $x^2 = 4y$: $4y + (y - r - 1)^2 = r^2$

D = 0 ⇒ r = 4

58) 

$A = \left(t^2, 2t\right); S = (1, 0); AC = BD = \frac{25}{4}$

Focal distance $AS = t^2 + 1; CS = \frac{21 - 4t^2}{4}$

$\frac{1}{CS} + \frac{1}{AS} = \frac{1}{a} \Rightarrow t = \pm 2, \pm\frac{1}{2}$

which are the points. Now find area.

59) sinα – sinβ + 4cosα cosβ

$2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} + 2\cos(\alpha+\beta) + 2\cos(\alpha-\beta)$

$\sin\left(\frac{\alpha-\beta}{2}\right) - 1 + 2\left[1 - 2\sin^2\left(\frac{\alpha-\beta}{2}\right)\right]$

$-\left[4\sin^2\left(\frac{\alpha-\beta}{2}\right) - \sin\left(\frac{\alpha-\beta}{2}\right) - 1\right]$

$-\left[\left(2\sin\left(\frac{\alpha-\beta}{2}\right) - \frac{1}{4}\right)^2 - \frac{17}{16}\right]$

range $\left[-4, \frac{17}{16}\right]$

 6 integers –4, –3, –2, –1, 0, 1

60) $\frac{1}{b} + \frac{1}{c} = \frac{1}{a+b+c} - \frac{1}{a}$

$\frac{c+b}{bc} = \frac{-(b+c)}{a(a+b+c)}$

$(b+c)\left(\frac{1}{bc} + \frac{1}{a(a+b+c)}\right) = 0$

⇒ b + c = 0 ⇒ b = – C

and $\frac{1}{bc} + \frac{1}{a(a+b+c)} = 0$

$\Rightarrow a^2 + a(b+c) + bc = 0$

$\Rightarrow (a+b)(a+c) = 0$

a = –b or a = –c

In any case expression is O.